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Dmytro RUMIANTSEV

Principal IT Consultant, MEng

ORCID: 0009-0007-1537-6683

bcomua@gmail.com

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AN INTEGRATED FRAMEWORK FOR AI-DRIVEN QUALITY ENGINEERING: SCALING TEST AUTOMATION THROUGH LARGE LANGUAGE MODELS AND BEHAVIOR-DRIVEN DEVELOPMENT

The purpose of this research is to develop, substantiate and validate an integrated framework for AI-driven Quality Engineering that enables scalable and accelerated test automation using Large Language Models (LLMs) in combination with the Behavior-Driven Development (BDD) paradigm. The work seeks to address the significant limitations of traditional testing methodologies within distributed software architectures and CI/CD pipelines, where engineering teams face substantial challenges in maintaining reliability while increasing delivery velocity. The aim is to create a replicable, practical model that increases automation coverage, reduces testing debt, restores confidence in failing legacy suites, and enables rapid extensibility of automated test assets.

Methodology. The research applies a qualitative case study methodology based on the implementation of an AI-assisted workflow within a large global connected services platform. The proposed framework combines MSpec (Machine Specifications) – a BDD-centric specification and testing language – with structured LLM prompting strategies and a multi-stage validation pipeline that ensures semantic correctness, deterministic consistency and executable reliability of AI-generated test code. Empirical evaluation included comparative measurement of time expenditures, diagnostic failure analysis, and validation of stability across large sets of integration tests. As a result, routine integration test creation time was reduced from approximately one working day to less than 30 minutes, while hundreds of legacy failing integration tests were successfully reconstructed and restored without manual rewriting.

The scientific novelty lies in the conceptualization of a hybrid operational Quality Engineering model, in which LLMs act as scalable automation accelerators, while human engineers remain responsible for cognitive oversight, architectural direction and strategic validation. The framework formalizes the operational fusion of LLM prompt engineering, MSpec-based specification languages, and layered machine validation – forming a repeatable, transferable, and industry-applicable model of AI-augmented test engineering.

Conclusions. The study empirically demonstrates that AI-assisted quality engineering has the capacity to radically increase test throughput, reduce regression backlog, improve maintainability and transform the economics of software testing in complex enterprise environments. The validated methodology provides organizations with a practical toolset to overcome the intrinsic constraints of legacy testing practices and meet modern system-level quality assurance demands at speed and scale.

Key words: quality assurance, test automation, large language models (LLMs), behavior-driven development (BDD), prompt engineering, machine specifications (MSPEC), software testing.

Дмитро РУМЯНЦЕВ

Головний IT-консультант, MEng

ORCID: 0009-0007-1537-6683

bcomua@gmail.com

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ІНТЕГРОВАНА РАМКА AI-ОРІЄНТОВАНОГО ІНЖЕНІРИНГУ ЯКОСТІ: МАСШТАБУВАННЯ АВТОМАТИЗАЦІЇ ТЕСТУВАННЯ ЗА ДОПОМОГОЮ ВЕЛИКИХ МОВНИХ МОДЕЛЕЙ ТА ПОВЕДІНКОВО-ОРІЄНТОВАНОЇ РОЗРОБКИ

Метою дослідження є розроблення, обґрунтування та емпірична валідація інтегрованої рамкової моделі AI-орієнтованого інженірингу якості, що забезпечує можливість масштабованої автоматизованої

генерації тестів шляхом поєднання великих мовних моделей (LLMs) та підходу розробки на основі поведінки (Behavior-Driven Development, BDD). Робота спрямована на подолання обмежень традиційних методів тестування в умовах розподілених архітектур та CI/CD-конвеєрів, коли команди розробників стикаються з труднощами підтримання надійності без зниження швидкості поставки програмного продукту. Метою є формування практичної та відтворюваної моделі, здатної збільшити охоплення тестуванням, зменшити «тестовий борг», підвищити стабільність існуючих тестових пакетів та прискорити розбудову автоматизованих тестових активів.

Методологія. Методологічною основою дослідження є якісний підхід кейс-стаді з апробацією AI-асистованого робочого процесу на прикладі великомасштабної глобальної платформи підключених сервісів. Запропонована модель інтегрує Machine Specifications (MSpec) – BDD-орієнтовану мову специфікацій – зі структурованими стратегіями формування запитів до LLM та багаторівневою процедурою валідації, що забезпечує коректність, узгодженість та виконуваність згенерованого тестового коду. Емпірична оцінка продемонструвала скорочення часу створення типових інтеграційних тестів з одного робочого дня до менш ніж 30 хвилин та успішне відновлення функціональності сотень існуючих помилкових тестів без їх ручної перебудови.

Наукова новизна проявляється у формуванні гібридної операційної моделі інжинірингу якості, за якої LLM виконують функцію масштабованих акселераторів автоматизації, а інженери забезпечують стратегічний контроль, семантичну валідацію та архітектурне спрямування. Запропонована рамкова модель формалізує інтеграцію інженерії підказок, BDD-орієнтованих мов специфікацій та багаторівневих механізмів машинної валідації, створюючи відтворювану та практично застосовну методику AI-підсиленої тестової інженерії.

Висновки. Отримані результати доводять, що використання штучного інтелекту в процесах забезпечення якості здатне суттєво підвищити пропускну здатність тестування, знизити регресійний беклог, покращити підтримуваність тестових пакетів та трансформувати економіку тестування у складних корпоративних середовищах. Запропонована методологія надає організаціям підтверджений практикою інструментарій для подолання обмежень традиційних тестових практик та забезпечення відповідності зростаючим вимогам системного контролю якості сучасних програмних продуктів.

Ключові слова: забезпечення якості, автоматизація тестування, великі мовні моделі (LLMs), розробка на основі поведінки (BDD), інженерія підказок (prompt engineering), машинні специфікації (MSpec), тестування програмного забезпечення.

Relevance of the Problem. The contemporary software development landscape is characterized by unprecedented acceleration in change velocity and an exponential rise in systemic complexity (Chen et al., 2021). The widespread adoption of microservice-based architectures, API-centric integrations, and Continuous Integration/Continuous Deployment (CI/CD) pipelines has created an increasingly demanding environment for Quality Assurance professionals (Farooq et al., 2023). Traditional testing strategies, which still rely heavily on manual processes and fragile automation frameworks, are failing to keep pace with modern development practices, resulting in severe bottlenecks in software delivery cycles. This imbalance between development velocity and testing capacity leads to the accumulation of “testing debt”, where outdated test suites, persistent failures, and insufficient regression coverage produce elevated risks to production system stability (Fowler & Lewis, 2014).

Recent advancements in artificial intelligence, particularly the emergence of highly capable Large Language Models (LLMs), have created new opportunities to address these challenges (Humble & Farley, 2010). While the potential of AI for transforming software testing is widely recognized, a substantial research gap remains in the development of practical, structured frameworks

for applying such technologies at enterprise scale. Much of the current scientific literature focuses on narrow or isolated use cases – for example, automated unit test generation – without offering a holistic methodology for integrating AI into complex, end-to-end quality assurance workflows (Mahida, 2021).

This paper aims to bridge this gap by proposing and evaluating a comprehensive framework that synergizes LLMs with the principles of Behavior-Driven Development (BDD). The core of this framework is the use of Machine Specifications (MSpec), a human-readable testing language, augmented by sophisticated prompt engineering techniques and a closed-loop validation process. The primary objective is to demonstrate how this integrated approach can systematically scale test automation, remediate extensive backlogs of failing tests, and ultimately restore trust in automated quality assurance processes. This research holds both theoretical significance, by outlining a new human-AI collaborative model for QA, and practical value, by offering an immediately applicable methodology for engineering teams.

Analysis of Recent Studies and Publications. The challenges of modern software testing are well-documented in both academic and industry literature. The transition towards distributed architectures has exponentially increased

the number of potential integration points, making exhaustive testing computationally infeasible (Malhotra, 2015). Traditional test design approaches frequently fail to cope with the complexity and non-deterministic behaviors inherent in microservice ecosystems. Simultaneously, CI/CD pipelines require rapid and continuous feedback, which legacy, brittle and often flaky automated test suites are not capable of providing efficiently (Nunan et al., 2020). As a result, engineering teams frequently ignore failing tests or disable entire segments of regression suites, creating the illusion of “100% pass rate stability” which masks latent quality risks (Fowler & Lewis, 2014).

In response to these shortcomings, AI-assisted testing has become an increasingly active research direction. Early implementations focused on machine learning techniques for defect prediction, using historical data to identify high-risk components that require prioritized test coverage (Malhotra, 2015). Further research has investigated the development of “self-healing” test automation, enabling AI algorithms to automatically adapt test scripts to minor changes in system interfaces or API contracts, thus reducing regression maintenance overhead (Shekhar, 2024). Commercial platforms such as Testim and Mabl operationalize these principles and offer AI-driven automation capabilities; however, such solutions frequently function as black boxes and cannot be easily adapted to bespoke enterprise systems or legacy testing frameworks.

Behavior-Driven Development (BDD) has been proposed as an approach for improving the maintainability and interpretability of test suites by grounding test specifications in natural language constructs that are accessible to both technical and non-technical stakeholders (Skeet, 2019). Frameworks such as Cucumber and SpecFlow have become widely adopted industry standards, enabling closer collaboration and ensuring alignment between implemented test logic and actual business requirements. The MSpec framework utilized in this study follows the same conceptual principles, structuring tests as readable behavioral scenarios that facilitate simpler maintenance and accelerate onboarding of new engineers (Tufano et al., 2019).

The most recent and disruptive development in this domain is the application of Large Language Models (LLMs), particularly those based on the GPT architecture, to software engineering tasks (Chen et al., 2021; Farooq et al., 2023). Prior research has demonstrated their capability in generating code, translating between programming languages, and identifying bugs. Several

studies have also explored the use of LLMs for automatically generating unit test cases from focal methods (Zimmermann et al., 2010). However, the application of LLMs to large-scale integration testing, legacy test remediation, and the systematic restoration of failing regression suites remains significantly underexplored. This gap leads to the central research problem addressed in this work: How can a structured framework that integrates LLM-assisted generation with BDD-based specification principles be implemented to effectively scale automated testing and eliminate accumulated testing debt within complex, enterprise-grade software systems?

The purpose of this research is to develop, substantiate and validate an integrated framework for AI-driven Quality Engineering that enables scalable and accelerated test automation using Large Language Models (LLMs) in combination with the Behavior-Driven Development (BDD) paradigm. The work seeks to address the significant limitations of traditional testing methodologies within distributed software architectures and CI/CD pipelines, where engineering teams face substantial challenges in maintaining reliability while increasing delivery velocity. The aim is to create a replicable, practical model that increases automation coverage, reduces testing debt, restores confidence in failing legacy suites, and enables rapid extensibility of automated test assets.

Presentation of the Main Research Material.

This research was conducted using a qualitative, single-case study methodology and supplemented by elements of action research, with the aim of developing and iteratively refining an AI-driven Quality Engineering framework under real enterprise conditions. The empirical environment of the study was a global-scale connected services platform characterized by highly distributed architecture and a large backlog of legacy, unstable integration tests – a context that is increasingly representative of contemporary software ecosystems (Malhotra, 2015).

The foundational methodological basis of the proposed framework is Behavior-Driven Development (BDD), a testing paradigm empirically shown to improve maintainability and clarity through the use of human-readable, natural language specifications (Skeet, 2019). Specifically, the Machine Specifications (MSpec) framework was selected due to its capacity to structure test cases as human-readable behavioral scenarios, which is essential for providing contextual clarity both for engineers and for the LLM-based generation mechanisms (Tufano et al., 2019). This approach directly addresses the challenges of onboarding

team members to complex projects and maintaining test suites over time.

An LLM was integrated into this BDD-oriented workflow as a co-processing partner to human engineers, performing tasks such as source code change analysis, generation of new integration test cases, and refactoring of fragile or obsolete test designs. This direction corresponds with recent empirical findings demonstrating the effectiveness of LLMs in executing sophisticated software engineering activities (Chen et al., 2021; Farooq et al., 2023). Interaction with the model was driven by strategic prompt engineering – a core control mechanism required to ensure semantically accurate and contextually valid outputs. Two distinct operational prompt patterns were defined within the framework: the first targeting new test generation to expand coverage, and the second specialized for legacy test remediation, where the model receives legacy code, updated classes, and specific error traces to derive a corrected implementation that resolves the failure scenario.

Recognizing the propensity of LLMs to generate output that may appear syntactically correct yet be semantically invalid or logically flawed, the framework incorporates a rigorous multi-stage validation protocol. This human-in-the-loop mechanism is critical for ensuring the reliability and production-suitability of AI-generated test code, an issue consistently emphasized in contemporary research on AI-assisted automation (Farooq et al., 2023). The protocol includes IDE-level static

analysis, iterative correction by feeding errors back to the LLM, runtime verification of the executed tests, and a final cross-prompt validation to ensure quality and accuracy. The primary limitation of this methodology is its reliance on a single case study, which may constrain the generalizability of the findings.

The implementation of the AI-driven Quality Engineering framework resulted in substantial and quantifiable improvements in testing throughput and overall delivery efficiency. The quantitative outcomes, summarized in Table 1, illustrate a dramatic acceleration in automated test creation relative to traditional manual workflows. For example, the time required to author a simple integration test decreased from approximately one working day to less than 30 minutes – representing an efficiency gain exceeding 90%. These improvements hold particular strategic relevance within CI/CD environments, where continuous, rapid feedback loops are required to sustain deployment velocity, yet traditional automation methods frequently fail to deliver such responsiveness (Nunan et al., 2020).

While complex, multi-service scenarios still required significant developer engagement and architectural reasoning, the AI-assisted workflow reduced the total effort by approximately half. This outcome is particularly important given the exponentially increasing complexity associated with microservice architectures, where the combinatorial explosion of integration paths makes traditional testing extremely resource-intensive (Malhotra, 2015).

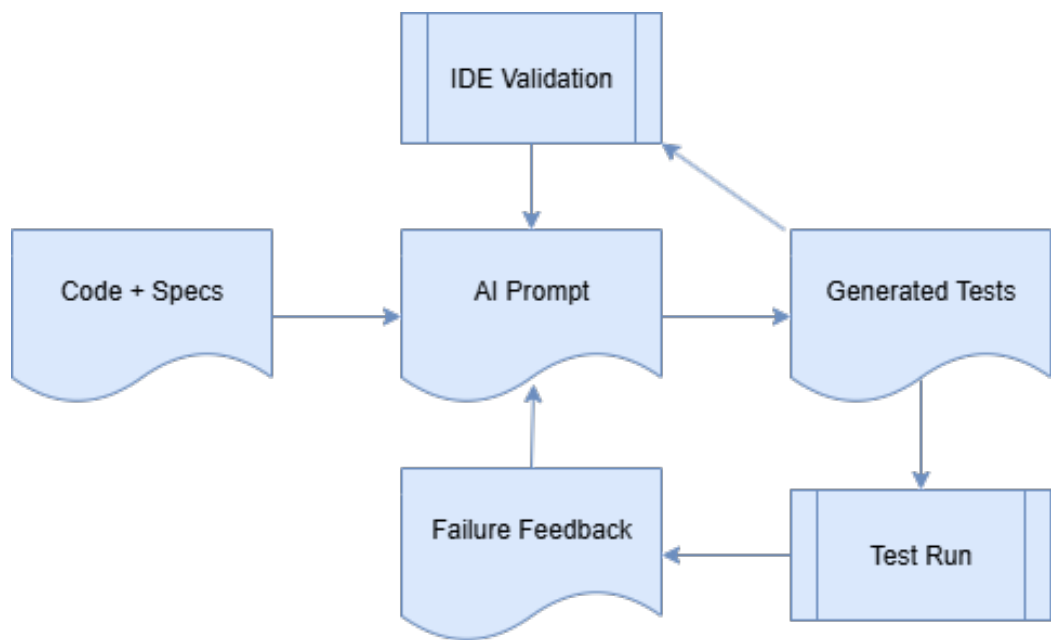


Fig. 1. The Closed-Loop Framework for AI-Assisted Quality Engineering

Table 1

Comparison of Manual vs. AI-Assisted Test Creation Effort

Scenario	Manual Effort	AI-Assisted Effort	Efficiency Gain
Simple Integration Test	~1 Day	~30 Minutes	>90%
Medium-Complexity Scenario	~4 Days	~1 Day	~75%
Complex Multi-Service Scenario	~2 Weeks	~1 Week	~50%

The most substantial impact was observed within the large-scale empirical case study, which had accumulated a pronounced “testing debt” over multiple years of continuous development (Fowler & Lewis, 2014). Using the legacy test remediation strategy, the project successfully restored 43 failing integration tests and their associated classes. The resulting fixes included addressing incomplete or inconsistent refactorings, introducing additional guardrails to prevent unstable edge-case behavior, and improving overall code maintainability. This directly counteracts the widely known industry phenomenon in which failing tests are routinely ignored, creating the illusion of “100% pass rates” while concealing systemic quality risks (Fowler & Lewis, 2014).

While the total runtime of the automated test suite experienced a marginal increase due to the reintroduction of functional tests, the defect detection rate remained stable throughout the observation period. The primary outcome of applying the framework was not only the resolution of existing technical failures but also the restoration of organizational trust in the automated testing pipeline as a dependable quality assurance gate. This revitalization of previously unusable testing assets provides a pragmatic solution for teams facing long-term maintenance burdens in legacy automation ecosystems — a challenge extensively highlighted in research related to automated fault prediction and AI-enabled automation practices (Ouyang et al., 2022; Shekhar, 2024).

The results of this study strongly support the hypothesis that a structured integration of LLMs within a BDD framework can substantially enhance the scalability and effectiveness of quality engineering. The transformative potential of this approach can be formally quantified using an Efficiency Gain (E_g) metric, defined as:

$$E_g (\%) = \frac{T_{\text{manual}} - T_{\text{AI}}}{T_{\text{manual}}} \times 100\%$$

Here, T_{manual} and T_{AI} represent the time for manual and AI-assisted efforts, respectively. The data from this study demonstrates E_g values exceeding 90% for routine tasks, a finding that highlights the profound impact of AI as a force multiplier for QA teams. This outcome is consistent with emerging research on the use of LLMs for code generation,

yet it extends prior work by introducing a practical and operationalizable workflow tailored to the significantly more complex domain of enterprise-scale integration testing (Zimmermann et al., 2010).

A critical interpretation of these findings is that the effectiveness of AI in Quality Engineering is driven not exclusively by the intrinsic capabilities of the model itself, but rather by the rigor and structure of the human–AI interaction framework. The synergy between MSPEC’s human-readable specification format, strategic prompt engineering, and the disciplined multi-stage validation process proved to be essential in achieving reliable outcomes. MSPEC provided the LLM with explicit semantic context, reducing ambiguity and ensuring alignment with business intent, while the prompt templates constrained model reasoning toward consistent and contextually correct solutions. The layered validation workflow acted as an essential safeguard against the introduction of flawed or logically incorrect AI-generated tests into production code. This directly responds to concerns repeatedly highlighted in the literature regarding the reliability and trustworthiness of AI-generated software artifacts (Farooq et al., 2023).

When compared with earlier research, this study provides a distinct contribution through its direct focus on remediation of accumulated testing debt. Whereas previous work has primarily examined the application of AI to generate new tests or to self-heal minor surface-level breakages in existing automation suites (Ouyang et al., 2022; Shekhar, 2024), this research demonstrates the capability of LLMs to analyze, interpret, and systematically repair large volumes of deeply flawed legacy integration tests. This represents a significantly more complex and high-impact operational scenario, since legacy remediation directly influences the long-term maintainability, stability, and trustworthiness of enterprise automated testing ecosystems. The practical implications for organizations can be further understood through the lens of Return on Investment (ROI), which connects the technical benefits to financial outcomes:

$$\text{ROI}(\%) = \frac{(C_h T_s) - C_{\text{impl}}}{C_{\text{impl}}} \times 100\%$$

In this model, the value derived from total hours saved (T_s) multiplied by the engineer’s hourly cost (C_h) is weighed against the implementation

cost (C_{impl}). This provides a formal justification for adopting AI-driven frameworks, demonstrating that the revitalization of critical testing assets is not only a technical necessity but also a financially sound strategy.

The theoretical implication of this work is the reframing of the QA engineer's role. In this new paradigm, the engineer transitions from a manual author of test scripts to an orchestrator of an AI-powered system. Their core competencies shift towards domain analysis, strategic prompt crafting, and critical validation of AI output. This hybrid model leverages the strengths of both parties: AI provides speed and scale for repetitive tasks, while human engineers provide the domain knowledge, critical thinking, and architectural oversight that AI currently lacks.

The limitations of this study, primarily its single-case nature, must be acknowledged. The results from one specific platform and technology stack may not be universally applicable without adaptation. Future research should seek to replicate this framework across different organizations, programming languages, and testing contexts to validate its broader applicability. Additionally, the rapid evolution of LLM technology means that the specific prompt patterns documented here may need to be refined as models become more capable. Further investigation into generative AI for creating exploratory test scenarios and integrating predictive analytics from production monitoring data could extend the boundaries of this framework.

Conclusions and Future Research Directions.

In an era where the pace of software delivery is relentless, AI is evolving from a peripheral technology to a core enabler of effective quality engineering. This research has presented and validated an integrated framework that synergizes Large Language Models with Behavior-Driven Development to address the critical industry challenge of scaling test automation. By embedding LLMs into a workflow anchored by Machine Specifications and disciplined human oversight, engineering teams can achieve dramatic reductions in the time required for test creation and maintenance, restore trust in neglected automation pipelines, and ultimately keep pace with the demands of modern software development.

The primary contribution of this work is a practical, replicable methodology that moves beyond theoretical discussions of AI in testing to offer a concrete set of strategies, including specific prompt patterns and validation protocols. The successful remediation of a massive backlog of failing tests in the case study demonstrates the framework's potent real-world impact.

The future of quality engineering lies in this hybrid approach, where AI is leveraged for its computational power and scale, and human developers provide the essential judgment, domain expertise, and strategic direction. Organizations that thoughtfully embrace this collaborative model, fostering a culture of data discipline and iterative adoption, will be best positioned to define the next generation of high-quality, reliable software delivery.

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